AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB)

Faculty of Science and Technology (FST)
Department of Mathematics
Undergraduate Program

COURSE PLAN	SEMESTER: SUMMER 2025-2026
I. Course Core and Title: MAT 2101: Complex Variables, Laplace and Z-transformations **II. Credit** 3 credit hours (3 hours of theory per week) **III. Nature** Science Core Course for CS, EEE **IV. Prerequisite:** MAT 1102: Differential Calculus and Coordinate Geometry, MAT 1205: Integral Calculus and ODE	**V. Vision:** Our vision is to be the preeminent Department of Mathematics through creating recognized professionals who will provide innovative solutions by leveraging contemporary research methods and development techniques of computing that are in line with the national and global context. **VI. Mission:** The mission of the Department of Mathematics of AIUB is to educate students in a student-centric dynamic learning environment; to provide advanced facilities for conducting innovative research and development to meet the challenges of the modern era of computing, and to motivate them towards a life-long learning process.

VII - Course Description:

- Explain Laplace transform, inverse Laplace transform and application of Laplace transform.
- Explain complex variables, its properties and complex algebra.
- Construct an analytic function.
- Explore complex integration using line integrals, Cauchy-residue theorem and Cauchy's integral formula.
- Apply Laurent's theorem to express the functions as series.
- Describe singularity, poles, zeros and residue of complex valued function.
- Explain Z-transform, inverse Z-transform and engineering and scientific applications of Z-transform.

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VIII – Course outcomes (CO) Matrix:

By the end of this course, students should be able to:

COs*	CO Description	Level of Domain***			PO Assessed ****
		C	P	A	
CO1	*Predict* and *perform* the basic concept of complex variables.	3			PO-a-3
CO2	*Compute* the Laplace- and Z- transformations for different functions and sequences.	3			PO-a-3
CO3 **	*Analyze* the functions to categorize, illustrate or differentiate.	4			PO-b-2
CO4 **	*Apply* Laplace- and Z-transformations to *solve* differential equations, difference equations and engineering problems.	3			PO-a-3

C: Cognitive; P: Psychomotor; A: Affective Domain

* *CO assessment method and rubric of COs assessment is provided in later section*

** *COs will be mapped with the Program Outcomes (POs) for PO attainment*
*** *The numbers under the 'Level of Domain' columns represent the level of Bloom's Taxonomy each CO corresponds to.*
**** *The numbers under 'PO Assessed' column represent the POs each CO corresponds to.*

IX – Topics to be covered in the classes*:

Week	Corresponding COs	Topic	Teaching-Learning Strategy	Assessment strategy
1	**CO1, CO2**	Definition, Application of Laplace Transformation, some important formulae and solving problems using those formulae. Definition of Unit Step Function, Rectangular Pulse, Laplace Transformation of Unit Step Function. Solving some problems associated with unit step function by Laplace Transformation.	Lecture delivery, Board work, Solving exercises, Discussion	Class Performance
2	**CO2, CO3**	Some important formulas, solving problems using direct formulae and property of inverse Laplace transformation. Inverse Laplace transformation using partial fraction (Denominator is formed by the linear factors, repeated factors, and quadratic factors). Inverse Laplace transformation of $F(s)$ with unit step function.	Lecture delivery, Board work, Solving exercises, Discussion	Quiz-1 on Lecture 4, Class Performance
3	**CO2, CO4**	Solving differential equations using Laplace transformation. Solving system of Ordinary Differential Equations (ODE) by Laplace Transform. Solution of some Engineering problems in RLC circuit, Mass-spring system etc..	Lecture delivery, Board work, Solving exercises, Discussion	Class Performance
4	**CO1, CO3**	Complex numbers, Graphical representation, Fundamental operations, Conjugates, Absolute value/modulus, Power of imaginary unit, Polar form and argument of complex number.	Lecture delivery, Board work, Solving exercises, Discussion	Quiz-2 On Lecture 7, Class Performance
5	**CO1, CO3**	Complex equations, complex roots and application of De Moivre's theorem. Mappings and Bilinear transformation. Complex function and its differentiation (Cauchy-Riemann equations).	Lecture delivery, Board work, Solving exercises, Discussion	Quiz-3 on Lecture 10, Class Performance
6	**CO1, CO3**	Application of Cauchy-Riemann conditions in physical problems in fluid flow, electrostatics, aerodynamics etc.	Lecture delivery, Board work, Solving exercises, Discussion	Class Performance
7	**Midterm Exam**			

8	**CO1, CO3**	Definition of line integral, path of integration, Evaluation of line integral along a simple curve, along a circle and curve consisting multiple paths.	Lecture delivery, Board work, Solving exercises, Discussion	Class Performance
9	**CO1, CO3**	Singularity: poles, zeros, and residues of a complex function, Cauchy Residue Theorem (CRT). Integral along a simple closed curve using CRT.	Lecture delivery, Board work, Solving exercises, Discussion	Quiz 1 on lecture 4, Class Performance
10	**CO3**	Types of improper integral $\int_{-\infty}^{\infty} f(x)\,dx$ using Jordan's Lemma and Cauchy's Residue Theorem. Laurent series.	Lecture delivery, Board work, Solving exercises, Discussion	Class Performance
11	**CO2, CO3, CO4**	Definition of Z-transform, its physical meaning and applications, Z-transform of some simple sequences, properties of Z-transform. Z-transform of discrete time unit step function and Kronecker delta function. Use of direct formulae of Z-transform for solving different problems.	Lecture delivery, Board work, Solving exercises, Discussion	Quiz-2 on Lecture 7, Class Performance
12	**CO2, CO4**	Inverse Z-Transform by Partial fraction and integral method. (Residue theorem).	Lecture delivery, Board work, Solving exercises, Discussion	Class Performance
13	**CO2, CO4**	Solution of linear difference equations by Z-transform. Solution of systems of difference equations by Z-transform. Solving some physical problems by Z-transform.	Lecture delivery, Board work, Solving exercises, Discussion	Quiz-3 on Lecture 11, Class Performance
14	**Final Exam**			

X – Mapping of PO to Courses and K, P, A

PO Indicator ID	PO Indicators Definition (As per the requirement of WKs)	Domain	K	P	A
PO-a-3	Apply information and concepts in engineering fundamentals to solve complex computer science and engineering problems with a range of conflicting requirements.	Cognitive Level 3 (Applying)	K3	P1 P2 P6	
PO-b-2	Formulate solutions, procedures, and methods using first principles of mathematics for engineering sciences.	Affective Level 4 (Responding)	K2		

XI – K, P, A Definitions

Indicator	Title	Description
K2	Conceptual based mathematics	Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline
K3	Theory based engineering fundamentals	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline
P1	Depth of knowledge required	Can not be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach
P2	Range of conflicting requirements	Involve wide-ranging or conflicting technical, engineering, and other issues
P6	Extent of stakeholder involvement and conflicting requirement	Involve diverse groups of stakeholders with widely varying needs

XII – Mapping of CO Assessment Method and Rubric

The mapping between Course Outcome(s) (COs) and The Selected Assessment method(s) and the mapping between Assessment method(s) and Evaluation Rubric(s) is shown below:

COs	Description	Mapped POs	Assessment Method	Assessment Rubric
CO1	Predict and perform the basic concept of complex variables.	PO-a-3	Quiz/Term Question & Assignment	Rubric for Quiz/Term Question & Assignment
CO2	Compute the Laplace- and Z- transformations for different functions and sequences.	PO-a-3	Quiz/Term Question & Assignment	Rubric for Quiz/Term Question & Assignment
CO3	Analyze the functions to categorize, illustrate or differentiate	PO-b-2	Quiz/Term Question & Assignment	Rubric for Quiz/Term Question & Assignment
CO4	Apply Laplace- and Z-transformations to solve differential equations, difference equations and engineering problems.	PO-a-3	Quiz/ Quiz/Term Question & Assignment	Rubric for Quiz/Term Question & Assignment

XIII – Evaluation and Assessment Criteria

CO1: *Predict* and *perform* the basic concept of complex variables.

Assessment Criteria	Not Attempted/ Incorrect (0)	Inadequate (1)	Satisfactory (2)	Excellent (3)
Fundamental Concept	No attempt to define or demonstrate understanding of the term.	Definition provided lacks clarity or accuracy, with limited or no examples.	The definition provided is clear and mostly accurate, with some examples.	Correctly and comprehensively define the term with examples.
Formulation	No attempt to provide formulae or identify techniques.	The formulae provided are incomplete or incorrect, or techniques are not appropriately identified.	Formulae provided is mostly correct but may lack completeness or clarity.	Write formulae to identify the appropriate technique.
Evaluation	No attempt to use De Moivre's theorem to find root of complex equation.	Attempt to use De Moivre's theorem, but with significant errors or incomplete steps.	Successfully use De Moivre's theorem using appropriate techniques, with minor errors.	Correct and comprehensive use of De Moivre's theorem.
Correctness of Answer	No attempt or incorrect answer provided, with no steps shown.	The answer provided contains notable errors, with some steps missing or incorrect.	Mostly correct answers provided, with few errors in calculation shown.	Arrived at correct answer with no error showing every step of calculation.

CO2: *Compute* the Laplace- and Z- transformations for different functions and sequences.				
Assessment Criteria	**Not Attempted/ Incorrect (0)**	**Inadequate (1)**	**Satisfactory (2)**	**Excellent (3)**
Fundamental Concept	No attempt or misunderstanding of Laplace and Z-transforms.	Identifies the transform to be used (Laplace or Z) but struggles to explain its purpose or characteristics.	Demonstrates a clear understanding of Laplace and Z-transforms and their relationship to continuous and discrete signals.	Identification of differential equations according to their order.
Formulation	No attempt or uses incorrect formulas.	Attempts to formulate the solution but uses partially incorrect or incomplete formulas.	Accurately formulates the solution using appropriate transformation formulas.	Write formulae to identify the appropriate techniques.
Evaluation	No attempt or applies the transformation incorrectly.	Makes some progress in applying the transformation but has significant errors in calculations or steps.	Applies the transformation with some minor errors or omissions in calculations.	Correctly finding Laplace and Z-transformation of different function and sequence.
Correctness of Answer	Incorrect final answer.	Partially correct answer, with significant errors or deviations from the expected result.	Mostly correct answers, with minor errors or omissions that do not significantly impact the result.	Arrived at correct answer with no error showing every step of calculation.

CO3: *Analyze* the functions to *categorize*, *illustrate* or *differentiate*.				
Assessment Criteria	**Not Attempted/ Incorrect (0)**	**Inadequate (1)**	**Satisfactory (2)**	**Excellent (3)**
Selection of Methodology	No attempt or choose completely inappropriate methods.	Selects some relevant methods but struggles to justify their choice or misses key aspects.	Choosing appropriate methods based on function type and analysis objective, demonstrating clear understanding of their application.	Identifies correct theory for relevant problems.
Formulation	No attempt or uses incorrect formulas/techniques.	Attempts to formulate the solution but uses partially incorrect or incomplete formulas/techniques.	Accurately formulates analysis using appropriate mathematical tools and techniques.	Write formulae to identify the appropriate techniques.
Evaluation	No attempt or applies methodology incorrectly, leading to significant errors.	Makes some progress in applying the methodology but has errors in calculations or interpretations.	Applies methodology with some minor errors or omissions but correctly interprets results.	Correctly checking analyticity of complex functions, finding line Integral, expanding a function in Laurent series, and solving complex equations.
Correctness of Answer	Incorrect conclusion or interpretation misrepresenting function's properties.	Partially correct answer with significant errors or deviations from the expected conclusion.	Mostly correct answers with minor errors or omissions that do not significantly impact the analysis.	Arrived at correct answer with no error showing every step of calculation.

CO4: *Apply* Laplace- and Z-transformations to *solve* differential equations, difference equation and engineering problems.				
Assessment Criteria	**Not Attempted/ Incorrect (0)**	**Inadequate (1)**	**Satisfactory (2)**	**Excellent (3)**
Fundamental Concepts	No attempt or misunderstanding of Laplace and Z-transforms, or differential/difference equations.	Identifies the type of equation (differential or difference) and the relevant transform (Laplace or Z) but struggles to explain their connection or purpose.	Demonstrates a clear understanding of mathematical concepts and their connection in problem-solving. Can identify the appropriate transform based on equation type and problem context.	Identification of difference equations according to their order.
Formulation	No attempt or uses incorrect formulas or methods.	Attempts to formulate the solution but uses partially incorrect or incomplete formulas/methods.	Accurately formulates the solution using appropriate transformation techniques, including initial/boundary conditions for differential equations.	Write formulae to identify the appropriate techniques.
Evaluation	No attempt or applies the transformation incorrectly, leading to significant errors.	Makes some progress in applying the transformation but has errors in calculations or interpretations.	Applies the transformation with some minor errors or omissions in calculations but arrives at a mostly correct solution.	Solving differential equations using Laplace-transformation and difference equations using Z-transformation.
Correctness of Answer	Incorrect final answer or solution that does not address the problem.	Partially correct answer with significant errors or deviations from the expected solution.	Mostly correct answers with minor errors or omissions that do not significantly impact the overall solution.	Arrived at correct answer with no error showing every step of calculation.

XIV- Course Requirements

- Students are expected to attend at least 80% of the class.
- Students are expected to participate actively in the class.
- For both terms, there will be 3 quizzes based on the theoretical knowledge and conceptual understanding of the topic covered discussed in the classes.

XV – Evaluation & Grading System*

Midterm and Final term	
Quiz	40%
Class Attendance	10%
Class Performance/Assignment	10%
Mid/ Final term Examination	40%
Total	100%
Final Grade/ Grand Total	
Midterm:	40%
Final Term:	60%

Letter	Grade Point	Numerical %
A+	4.00	90-100
A	3.75	85 - < 90
B+	3.50	80 - < 85
B	3.25	75 - < 80
C+	3.00	70 - < 75
C	2.75	65 - < 70
D+	2.50	60 - < 65
D	2.25	50 - < 60
F	0.00	< 50
I		Incomplete
W		Withdrawal
UW		Unofficially Withdrawal

** The evaluation system will be strictly followed as per the AIUB grading policy.*

** CO attainment will be achieved with 60% of the evaluation marks.*

** A 'UW' grade is assigned when a student is unable to meet course requirements, like inadequate attendance (at least 80% of the classes for each of their courses for the semester.), not appearing in quizzes and/or term exams, not submitting assignments.*

XVI – Textbook/ References

1. Advanced Engineering Mathematics (10th edition) by Erwin Kreyszig, Herbert Kreyszig, Edward J. Norminton, published by John Wiley & Sons, Inc.
2. Complex Variables and Applications – R.V. Churchill and J.W.Brown.
3. Laplace Transform – Murray R. Spiegel (Schaum's Outline Series)
4. Complex Variables and Applications M.R.Spiegel (Schaum's Outline Series).
5. Recurrence Relations in Teaching Students of Informatics by Valentin P. BAKOEV.
6. *Z*-Transform and Its Application to Development of Scientific Simulation Algorithms by ZVONKO FAZARINC, Compute Appl Eng Educ:21:75-88,2013.
7. Lecture Notes.

XVII - List of Faculties Teaching the Course

FACULTY NAME	

XVIII – Verification

Prepared by:	Moderated by:	Checked by:
---------- **Roushanara Begum** *Course Convener* ---------- **Khadiza Akter Mitu** *Coordinator* ---------- **Zasmin Hauqe** *Coordinator* Date:.....	---------- **Dr. M. Mahmudul Hasan** *Point Of Contact* *OBE Implementation Committee* Date:.....	---------- **Dr. M. Mostafizur Rahman** *Head, Department of Mathematics* Date:.....
Verified by:	**Certified by:**	**Approved by:**
..... **Dr. Md. Abdullah-Al-Jubair** *Director* *Faculty of Science & Information Technology* Date:.....	..... **Prof. Dr. Dip Nandi** *Associate Dean,* *Faculty of Science & Information Technology* Date:.....	..... **Mr. Mashiour Rahman** *Dean,* *Faculty of Science & Information Technology* Date:.....